

Crystal-Controlled Shortwave Transmitter

Pradeep Vasudeva

A shortwave transmitter can be built with IC CD4049, which is used as an oscillator and a buffer amplifier. The transmitter can be crystal controlled to work at a frequency of 3MHz to 5MHz, which includes the ham radio bands of 3.5MHz to 3.7MHz and 3.89MHz to 3.9MHz. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the shortwave transmitter using CD4049 and 4MHz crystal is shown in Fig. 2. It is built around LM386 (IC1), CD4049 (IC2), and transistor BC547 (T1).

The IC CD4049 is a hex inverter that has six NOT or inverter gates of which only four are used in this circuit. The inverter gate between pin numbers 7 and 6 is used as a crystal controlled oscillator.

The output of the oscillator is given to the inverter gate between pin numbers 5 and 4, which is used as a buffer. The buffer's output goes to two inverter gates between pin numbers 9 and 10 and 11 and 12, which are connected in parallel. The output of these gates goes to transistor T1, which is used as a modulator as well as class C amplifier.

This method of modulation is different from the modulation techniques used in transmitter circuits published earlier in EFY and is known as power conversion. If a class C device amplifying one frequency has its input power varied at second frequency, the output is modulated at the second frequency.

IC LM386 in the circuit is used as an audio frequency amplifier. The output of this amplifier goes to the collector of transistor T1 through the tuned circuit comprising capacitor C5 and inductor L1. It should be noted

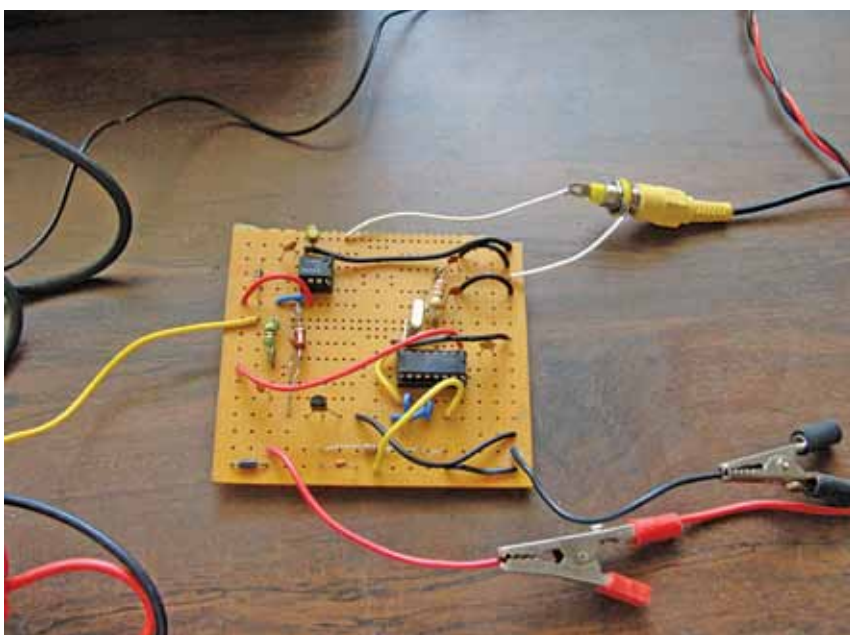


Fig. 1: Author's prototype

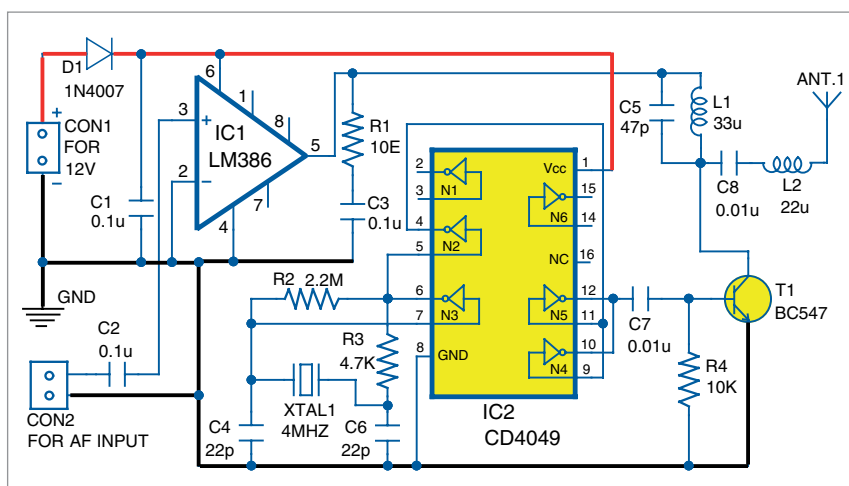


Fig. 2: Circuit diagram of shortwave transmitter

that the output of LM386 goes directly to the transistor and not through a capacitor, which is done when the IC is used as an audio frequency amplifier to drive a loudspeaker.

The DC voltage at collector of transistor should be half the power supply voltage under no-signal conditions. Thus, power to the transistor is varied at audio frequency and the amplitude